

CBSE Class 12 Chemistry — Half-Yearly Predicted Paper (Set 3 of 10)

Section A: 1-Mark Questions (MCQs / Assertion-Reason)

- Q1.** The unit of ebullioscopic constant (K_b) is: (a) $K\ kg\ mol^{-1}$ (b) $mol\ kg\ K^{-1}$ (c) $kg\ mol^{-1}\ K^{-1}$ (d) $K\ mol\ kg^{-1}$ [Solutions] [PYQ 2014] [Confidence: Very High]
- Q2.** Which of the following cells is commonly used in hearing aids and watches? (a) Dry cell (b) Lead storage battery (c) Mercury cell (d) Fuel cell [Electrochemistry] [PYQ 2017] [Confidence: Very High]
- Q3.** For a chemical reaction, $Rate = k[A]^1[B]^2$. If the volume of the reaction vessel is suddenly reduced to one-third of its original volume, the rate of the reaction will become: (a) 9 times (b) 27 times (c) 3 times (d) 1/27 times [Chemical Kinetics] [Practice] [Confidence: High]
- Q4.** Which of the following transition elements exhibits only a +3 oxidation state? (a) Manganese (Mn) (b) Iron (Fe) (c) Scandium (Sc) (d) Titanium (Ti) [The d- and f-Block Elements] [PYQ 2020] [Confidence: Very High]
- Q5.** The denticity of the ligand EDTA (Ethylenediaminetetraacetate) is: (a) Bidentate (b) Tridentate (c) Tetradentate (d) Hexadentate [Coordination Compounds] [CBSE PDF] [Confidence: Very High]
- Q6.** The Swarts reaction is specifically used for the preparation of: (a) Alkyl chlorides (b) Alkyl bromides (c) Alkyl fluorides (d) Alkyl iodides [Haloalkanes and Haloarenes] [PYQ 2015] [Confidence: Very High]
- Q7.** The controlled oxidation of a primary alcohol using Pyridinium Chlorochromate (PCC) yields: (a) A carboxylic acid (b) A ketone (c) An aldehyde (d) An ester [Alcohols, Phenols and Ethers] [PYQ Term-1 2021] [Confidence: Very High]
- Q8.** Which of the following compounds will undergo the Hell-Volhard-Zelinsky (HVZ) reaction? (a) $HCHO$ (b) CH_3COOH (c) C_6H_5COOH (d) CH_3COCH_3 [Aldehydes, Ketones and Carboxylic Acids] [PYQ 2021] [Confidence: Very High]
- Q9.** Two different solutions having the same molar concentration at the same temperature will invariably have the same: (a) Vapor pressure (b) Boiling point (c) Osmotic pressure (d) None of the above [Solutions] [Delhi Gov. SQP 2021] [Confidence: High]
- Q10.** The half-life period of a first-order reaction is: (a) Directly proportional to the initial concentration (b) Inversely proportional to the initial concentration (c) Independent of the initial concentration (d) Proportional to the square of the initial concentration [Chemical Kinetics] [Practice] [Confidence: Very High]
- Q11.** The general electronic configuration of transition elements is: (a) $(n-1)d^{1-10}ns^{1-2}$ (b) $(n-1)d^{10}ns^2$ (c) $(n-1)d^{1-10}ns^2$ (d) $(n-1)d^5ns^{1-2}$ [The d- and f-Block Elements] [Practice] [Confidence: High]
- Q12.** Which of the following isomers of bromobutane is a chiral molecule? (a) 1-Bromobutane (b) 2-Bromobutane (c) 1-Bromo-2-methylpropane (d) 2-Bromo-2-methylpropane [Haloalkanes and Haloarenes] [PYQ 2014] [Confidence: Very High]
- Directions for Q13 to Q16:** A statement of Assertion (A) followed by a statement of Reason (R) is given. Choose the correct answer out of the following choices: (a) Both A and R are true and R is the correct explanation of A. (b) Both A and R are true but R is not the correct explanation of A. (c) A is true but R is false. (d) A is false but R is true.

Q13. Assertion (A): The conductivity of an electrolytic solution decreases with a decrease in concentration (dilution). Reason (R): The number of ions per unit volume that carry the current in a solution decreases on dilution. [Electrochemistry] [PYQ 2015] [Confidence: Very High]

Q14. Assertion (A): Phenol undergoes electrophilic substitution reactions more readily than benzene, and the substitution occurs mainly at ortho and para positions. Reason (R): The $-\text{OH}$ group in phenol is ring activating and ortho-para directing due to the +R (resonance) effect. [Alcohols, Phenols and Ethers] [PYQ 2015] [Confidence: Very High]

Q15. Assertion (A): The complex $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$ is strongly paramagnetic. Reason (R): H_2O is a weak field ligand, so it does not cause forced pairing of electrons in the $3d$ orbitals of Fe^{3+} . [Coordination Compounds] [PYQ 2016] [Confidence: High]

Q16. Assertion (A): Formic acid (HCOOH) gives a positive Tollen's test, unlike other carboxylic acids. Reason (R): Formic acid contains both an aldehydic group and a carboxyl group in its structure. [Aldehydes, Ketones and Carboxylic Acids] [Practice] [Confidence: Medium]

Section B: 2-Mark Questions (Short Answer)

Q17. State Raoult's law for a solution containing non-volatile solutes. How does it differ conceptually from Henry's law? [Solutions] [PYQ 2014] [Confidence: Very High]

Q18. Explain the following variations in electrochemistry: (a) Why does the molar conductivity (Λ_m) of a strong electrolyte increase slowly with dilution? (b) Why cannot the limiting molar conductivity (Λ_m°) of a weak electrolyte be determined accurately by extrapolating the Λ_m vs $\sqrt{c}$ graph? [Electrochemistry] [PYQ Term-2 2022] [Confidence: Very High]

Q19. Draw the structures of the optical isomers (enantiomers) of the complex $[\text{Co}(\text{en})_3]^{3+}$. [Coordination Compounds] [PYQ 2015] [Confidence: Very High]

Q20. Account for the following: (a) The dipole moment of chlorobenzene is significantly lower than that of cyclohexyl chloride. (b) Alkyl halides, though polar, are completely immiscible with water. [Haloalkanes and Haloarenes] [PYQ 2016 / 2017] [Confidence: Very High]

Q21. Write the balanced chemical equations involved in the following named reactions: (a) Kolbe's reaction (b) Williamson ether synthesis [Alcohols, Phenols and Ethers] [PYQ 2023 / 2014] [Confidence: Very High]

Section C: 3-Mark Questions (Short Answer)

Q22. 15.0 g of an unknown substance dissolved in 450 g of water depressed the freezing point by 0.34 K. Calculate the molar mass of the unknown substance. (Given: K_f for water = $1.86 \text{ K kg mol}^{-1}$). [Solutions] [Practice] [Confidence: High]

Q23. A chemical reaction is first order with respect to reactant **A** and second order with respect to reactant **B**. (a) Write the differential rate equation for the reaction. (b) How is the rate of reaction affected if the concentration of **B** is tripled? (c) How is the rate affected if the concentrations of both **A** and **B** are doubled? [Chemical Kinetics] [PYQ 2016] [Confidence: Very High]

Q24. Account for the following observations in d-block chemistry: (a) Cerium (IV) (Ce^{4+}) is widely used as a good oxidizing agent in analytical chemistry. (b) Transition elements show significant catalytic properties. (c) The copper(I) ion (Cu^+) is unstable in aqueous solutions and undergoes disproportionation. [The d- and f-Block Elements] [PYQ 2023 / 2022] [Confidence: Very High]

Q25. How will you successfully carry out the following organic conversions in not more than two steps? (a) Chlorobenzene to biphenyl (b) Propene to 1-iodopropane (c) Ethanol to but-1-yne [Haloalkanes and Haloarenes] [PYQ 2016 / Practice] [Confidence: Very High]

Q26. Write the detailed mechanism (using curved arrows) for the acid-catalysed hydration of ethene to yield ethanol. [Alcohols, Phenols and Ethers] [PYQ 2019] [Confidence: Very High]

Q27. Give simple, observable chemical tests to distinguish between the following pairs of compounds: (a) Formic acid and Acetic acid (b) Acetophenone and Benzophenone (c) Ethanal and Propanone [Aldehydes, Ketones and Carboxylic Acids] [PYQ 2014 / Practice] [Confidence: Very High]

Q28. An organic compound 'A' with the molecular formula C_3H_6O is heavily resistant to oxidation but forms a secondary alcohol 'B' on catalytic reduction. Compound 'B' on heating with concentrated H_2SO_4 at 443 K undergoes dehydration to form a gas 'C' with the formula C_3H_6 . Identify the exact structures and IUPAC names of compounds A, B, and C. [Aldehydes, Ketones and Carboxylic Acids / Alcohols] [PYQ 2016 / Practice] [Confidence: High]

Section D: 4-Mark Questions (Case-Study)

Q29. Case Study: Commercial Electrochemical Cells Batteries and fuel cells are practical applications of galvanic cells. A secondary battery, such as the lead storage battery, is rechargeable and is heavily used in automobiles and inverters. It consists of a lead anode and a grid of lead packed with lead dioxide (PbO_2) as the cathode. A 38% solution of sulfuric acid acts as the electrolyte. When the battery is in use, it acts as a galvanic cell, and when it is being recharged, it acts as an electrolytic cell. (a) Write the balanced chemical reaction occurring at the anode during the discharging of a lead storage battery. (1 Mark) (b) What happens to the concentration (and density) of the sulfuric acid electrolyte during the discharging process? (1 Mark) (c) Write the overall cell reaction during the recharging of the battery. Does recharging require a DC or AC external voltage? (2 Marks) [Electrochemistry] [CBSE Term-2 2022 / Practice] [Confidence: High]

Q30. Case Study: The Arrhenius Equation and Activation Energy The dependence of the rate constant of a reaction on temperature is elegantly expressed by the Arrhenius equation: $k = Ae^{-E_a/RT}$. According to collision theory, a reaction occurs when reactant molecules collide with sufficient kinetic energy (activation energy, E_a) and proper orientation. The presence of a catalyst also profoundly impacts the reaction rate by providing an alternative pathway. (a) Define the term 'Activation Energy' (E_a). (1 Mark) (b) How does the addition of a positive catalyst increase the rate of a reaction in terms of the activation energy? (1 Mark) (c) The rate of a chemical reaction triples when the temperature is increased from 300 K to 310 K . Write the specific Arrhenius formula utilized to calculate the activation energy for two different temperatures, and substitute these given values into it. (Calculation of final E_a is not mandatory). (2 Marks) [Chemical Kinetics] [PYQ 2015 / Practice] [Confidence: Very High]

Section E: 5-Mark Questions (Long Answer)

Q31. (a) Define Osmotic pressure. Why is the osmotic pressure measurement preferred over other colligative properties for determining the molar masses of macromolecules (like proteins and polymers)? (2 Marks) (b) A solution containing 8.0 g of an unknown non-volatile substance dissolved in 100 g of diethyl ether boils at 36.86°C . If pure diethyl ether boils at 35.60°C , determine the molar mass of the unknown solute. (Given: Ebullioscopic constant K_b for diethyl ether = $2.02\text{ K kg mol}^{-1}$). (3 Marks) [Solutions] [Practice / PYQ themes] [Confidence: Very High]

Q32. (a) Write the balanced ionic equations for the following important redox reactions: (i) Oxidation of Fe^{2+} to Fe^{3+} by MnO_4^- in an acidic medium. (ii) Reaction of $Cr_2O_7^{2-}$ with H_2S in an acidic medium. (2 Marks) (b) Using Crystal Field Theory (CFT), explain why the complex $[Ti(H_2O)_6]^{3+}$ is coloured in an aqueous solution. What will happen to the colour of the complex if it is heated to completely remove the water ligands? (2 Marks) (c) Write the IUPAC name of the complex: $[Co(NH_3)_5(CO_3)]Cl$. (1 Mark) [The d- and f-Block Elements / Coordination Compounds] [PYQ 2017 / 2015] [Confidence: Very High]

Q33. (a) An aromatic organic compound 'A' with the molecular formula $C_9H_{10}O$ forms a 2,4-DNP derivative, reduces Tollen's reagent, and undergoes the Cannizzaro reaction. On vigorous oxidation with acidic $KMnO_4$, it gives 1,2-benzenedicarboxylic acid (Phthalic acid). Deduce the exact structure of compound 'A' and write the chemical equations for its reaction with Tollen's reagent and its Cannizzaro reaction. (3 Marks) (b) Write short notes describing the reagents and the transformation occurring in the following named reactions: (i) Stephen reduction (1 Mark) (ii) Decarboxylation of carboxylic acids (1 Mark) [Aldehydes, Ketones and Carboxylic Acids] [PYQ 2015 / Practice] [Confidence: High]

Answer Key

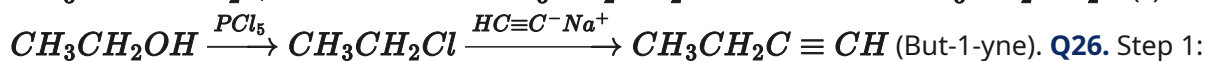
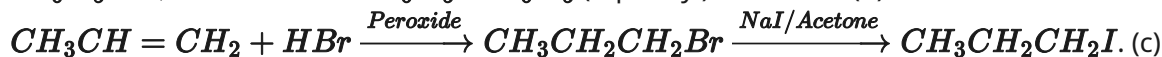
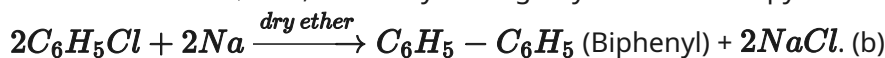
Q1. (a) $K\ kg\ mol^{-1}$. **Q2.** (c) Mercury cell. **Q3.** (b) 27 times. (Volume reduced to 1/3 means concentration of both A and B becomes 3 times. Rate = $k[3A]^1[3B]^2 = 27 \times k[A][B]^2$). **Q4.** (c) Scandium (Sc). (It loses its three outer electrons to form a stable noble gas configuration $3d^0$). **Q5.** (d) Hexadentate. (It binds through two nitrogen and four oxygen atoms). **Q6.** (c) Alkyl fluorides. (Reacting alkyl chloride/bromide with metallic fluorides like AgF). **Q7.** (c) An aldehyde. (PCC stops oxidation at the aldehyde stage). **Q8.** (b) CH_3COOH . (It has an alpha-hydrogen atom adjacent to the carboxyl group). **Q9.** (c) Osmotic pressure. ($\pi = CRT$. If concentration C and temp T are same, π is same). **Q10.** (c) Independent of the initial concentration. ($t_{1/2} = \frac{0.693}{k}$). **Q11.** (a) $(n-1)d^{1-10}ns^{1-2}$. **Q12.** (b) 2-Bromobutane. (The carbon carrying the bromine is bonded to four different groups: H, Br, Methyl, and Ethyl). **Q13.** (a) Both A and R are true and R is the correct explanation of A. **Q14.** (a) Both A and R are true and R is the correct explanation of A. **Q15.** (a) Both A and R are true and R is the correct explanation of A. (Weak field ligand means $\Delta_o < P$, causing $3d^5$ to have 5 unpaired electrons). **Q16.** (a) Both A and R are true and R is the correct explanation of A. (Formic acid $H-COOH$ acts as an aldehyde on the left side).

Q17. Raoult's law states that the relative lowering of vapor pressure of a solution containing a non-volatile solute is equal to the mole fraction of the solute. Henry's Law applies to gases dissolved in liquids ($P = K_H \cdot x$). Raoult's law is a special case of Henry's law where K_H becomes equal to the vapor pressure of the pure solvent (P°). **Q18.** (a) Dilution decreases ionic interactions and increases ionic mobility slightly, leading to a slow increase in Λ_m . (b) For a weak electrolyte, the degree of dissociation (α) increases steeply at extremely low concentrations, making the curve asymptotic to the y-axis, making accurate extrapolation impossible. **Q19.** Draw the 'propeller' shaped octahedral structures for dextro (d) and laevo (l) forms of $[Co(en)_3]^{3+}$ which are non-superimposable mirror images of each other. **Q20.** (a) In chlorobenzene, the C-Cl bond is attached to an sp^2 hybridized carbon, which is more electronegative than the sp^3 carbon in cyclohexyl chloride, reducing the bond's polarity and length. (b) Alkyl halides cannot form hydrogen bonds with water molecules, nor can they break the existing strong H-bonds in water. **Q21.** (a) Kolbe's Reaction: Phenol + $NaOH \rightarrow$ Sodium Phenoxide. Sodium Phenoxide + $CO_2/H^+ \xrightarrow{400K, 4-7atm}$ Salicylic Acid. (b) Williamson Synthesis: $R-X + R'-O^-Na^+ \rightarrow R-O-R' + NaX$.

Q22. $\Delta T_f = K_f \times \frac{W_B \times 1000}{M_B \times W_A}$. $0.34 = 1.86 \times \frac{15 \times 1000}{M_B \times 450}$. $M_B = \frac{1.86 \times 15000}{0.34 \times 450} = \frac{27900}{153} = 182.35\ g\ mol^{-1}$.

Q23. (a) Rate = $k[A]^1[B]^2$. (b) Tripling B: Rate' = $k[A]^1[3B]^2 = 9 \times k[A][B]^2$. Rate increases by 9 times.

(c) Doubling A & B: Rate = $k[2A]^1[2B]^2 = 2 \times 4 \times k[A][B]^2 = 8 \times \text{Rate}$. Rate increases 8 times. **Q24.** (a) Ce^{4+} is unstable and easily gains an electron to reach the highly stable +3 state characteristic of lanthanoids, thus acting as an oxidizer. (b) They provide vacant d-orbitals and exhibit variable oxidation states, forming unstable intermediates with reactants. (c) Cu^+ disproportionates in water: $2\text{Cu}^+ \rightarrow \text{Cu}^{2+} + \text{Cu}$, driven by the high hydration enthalpy of Cu^{2+} . **Q25.** (a) Fittig reaction:



Protonation of alkene: $\text{CH}_2 = \text{CH}_2 + \text{H}^+ \rightleftharpoons \text{CH}_3 - \text{CH}_2^+$. Step 2: Nucleophilic attack of water on carbocation: $\text{CH}_3 - \text{CH}_2^+ + \text{H}_2\text{O} \rightleftharpoons \text{CH}_3 - \text{CH}_2 - \text{OH}_2^+$. Step 3: Deprotonation to form ethanol: $\text{CH}_3 - \text{CH}_2 - \text{OH}_2^+ \rightarrow \text{CH}_3 - \text{CH}_2 - \text{OH} + \text{H}^+$. **Q27.** (a) Tollen's test: Formic acid gives a silver mirror, acetic acid does not. (b) Iodoform test: Acetophenone gives a yellow precipitate of CHI_3 with NaOH/I_2 , benzophenone does not. (c) Iodoform test: Ethanal gives a yellow precipitate (CHI_3), Propanone also does! *Correction:* Use Tollen's Test. Ethanal (aldehyde) gives a silver mirror, Propanone (ketone) does not. **Q28.** A is Propanone (Acetone) - a ketone, resistant to oxidation. CH_3COCH_3 . B is Propan-2-ol - a secondary alcohol. $\text{CH}_3\text{CH}(\text{OH})\text{CH}_3$. C is Propene - an alkene formed by dehydration. $\text{CH}_3\text{CH} = \text{CH}_2$.

Q29. (a) Anode discharging: $\text{Pb(s)} + \text{SO}_4^{2-}(\text{aq}) \rightarrow \text{PbSO}_4(\text{s}) + 2\text{e}^-$. (b) Sulfuric acid is consumed, so its concentration and density decrease. (c) Recharging reaction:

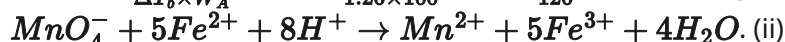
$2\text{PbSO}_4(\text{s}) + 2\text{H}_2\text{O(l)} \rightarrow \text{Pb(s)} + \text{PbO}_2(\text{s}) + 2\text{H}_2\text{SO}_4(\text{aq})$. It requires an external Direct Current (DC) voltage greater than the cell's potential. **Q30.** (a) The minimum extra amount of energy required by reacting molecules to form the activated complex and yield products. (b) A catalyst lowers the activation energy (E_a) barrier, providing an alternative pathway, so more molecules have enough energy to react.

(c) $\log\left(\frac{k_2}{k_1}\right) = \frac{E_a}{2.303R} \left(\frac{T_2 - T_1}{T_1 T_2}\right) \Rightarrow \log(3) = \frac{E_a}{2.303 \times 8.314} \left(\frac{310 - 300}{300 \times 310}\right)$.

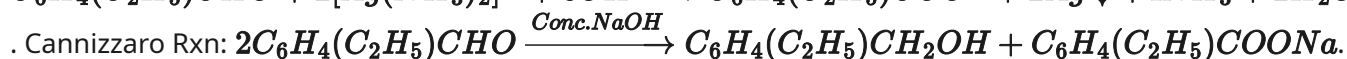
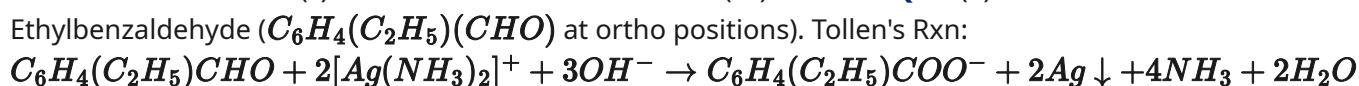
Q31. (a) The extra pressure that must be applied to the solution side to stop the flow of solvent molecules through a semi-permeable membrane. It is preferred because it can be measured at room temperature, and the magnitude of π is large even for very dilute solutions of macromolecules. (b)

$$\Delta T_b = 36.86 - 35.60 = 1.26 \text{ K (or } ^\circ\text{C)}.$$

$$M_2 = \frac{K_b \times W_B \times 1000}{\Delta T_b \times W_A} = \frac{2.02 \times 8.0 \times 1000}{1.26 \times 100} = \frac{16160}{126} \approx 128.25 \text{ g mol}^{-1}. \text{ Q32. (a) (i)}$$



$\text{Cr}_2\text{O}_7^{2-} + 3\text{H}_2\text{S} + 8\text{H}^+ \rightarrow 2\text{Cr}^{3+} + 3\text{S} + 7\text{H}_2\text{O}$. (b) Ti^{3+} has a $3d^1$ configuration. In the presence of H_2O ligands, d-orbitals split. The unpaired electron undergoes d-d transition absorbing visible light, imparting colour. If heated, ligands leave, crystal field splitting is destroyed, no d-d transition occurs, and it becomes colourless. (c) Pentaamminecarbonatocobalt(III) chloride. **Q33.** (a) Structure of A: 2-Ethylbenzaldehyde ($\text{C}_6\text{H}_4(\text{C}_2\text{H}_5)(\text{CHO})$ at ortho positions). Tollen's Rxn:



(b) (i) Stephen reduction: Nitriles are reduced to imines using SnCl_2 and HCl , which on hydrolysis give aldehydes. (ii) Decarboxylation: Sodium salts of carboxylic acids when heated with soda lime ($\text{NaOH} + \text{CaO}$) lose CO_2 to form alkanes containing one carbon less than the acid.